Towards a Serverless Intelligent Firewall: Integrating Cross-Cloud Adaptation, AI-Driven Security, and Zero-Trust Architectures

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***Abstract*—Businesses still have to deal with identity sprawl, fragmented cloud management, and cloud-native attack vectors that bypass conventional perimeter defences. This paper presents SIF-CCA, a serverless intelligent firewall that integrates a unified Zero-Trust Control Plane, multi-cloud serverless orchestration, BiGRU-driven temporal modelling, and XGBoost-based feature learning. The suggested model obtained 98% accuracy with an AUC-ROC of 0.990 on a thoroughly preprocessed CIC-IDS2017 dataset that included redundant feature removal, anomaly correction, attack consolidation, outlier filtering, and balanced class sampling. It performs better than standalone XGBoost, CNN, Random Forest, and LSTM. Cross-cloud experiments across AWS, Azure, and Google Cloud demonstrate low-latency intrusion detection at 135 ms, linear scalability under heavy workloads, and cost-efficient serverless execution. Additionally, a zero-trust evaluation shows up to 99.6% policy consistency, with low propagation latency and identity verification, proving that the architecture is appropriate for distributed environments. A gap analysis was conducted to identify what is missing from current solutions: serverless-native, end-to-end multi-cloud zero-trust solutions with quantifiable SLOs for latency-sensitive security tasks. SIF-CCA bridges this gap by offering adaptive AI-driven analytics, cloud-agnostic identity and policy abstraction, and automated threat response, hence reducing detection and recovery times by maintaining operational agility.**

***Index Terms*—Zero Trust, SIF-CCA, Serverless Security, Multi-Cloud, AI-Driven Detection, Cloud Orchestration, Intrusion Detection.**

I. Introduction

The deployment, management, and security of mission-critical applications are being altered by the growing dependence on multi-cloud and serverless architecture. Traditional perimeter-based security models are unable to handle the reality of identity sprawl, uneven policy enforcement, and dynamic cloud-native attack vectors as businesses spread workloads across AWS, Azure, and Google Cloud. Ensuring trustworthy, resilient, and adaptive security across heterogeneous platforms has therefore become a central challenge in modern cloud computing. Recent works present important advances, yet they equally highlight some significant limitations. Earlier defect-identification frameworks, for example, improve the reliability of health informatics systems but do not go so far as to secure large-scale, cloud-native workloads [1]. AI-empowered risk management frameworks now reinforce predictive capabilities, yet they remain domain-specific and without mechanisms for cross-cloud enforcement [2]. DevOps-SecOps relations create secure analytics pipelines but still do not leverage real-time intrusion detection or multi-cloud adaptive response capabilities. Despite the fact that zerotrust models provide better assurance about identity and access, they are broadly limited to vendor-specific restrictions, interoperability gaps, and performance trade-offs in a distributed system [3], [4]. AI-enabled orchestration engines automate workload placement and policy enforcement, but they stop at integrating ML-driven threat detection or unified Zero-Trust governance [5]. In addition, microsegmentation and continuous adaptive risk assessment frameworks provide fine-grained control but lack serverless-native scalability and end-to-end multi-cloud coordination [6].

These restrictions point to a key research gap: the pressing need for a unified, serverless, AI-driven security architecture providing automated attack responses and consistent Zero-Trust enforcement across heterogeneous cloud providers. In light of the above gap, this paper proposes SIF-CCA, a serverless intelligent firewall for Cross-Cloud Adaptation that includes hybrid machine learning-based detection, event-driven serverless orchestration, and a unified Zero-Trust Control Plane. The aim is to provide a cloud-agnostic, scalable, and real-time security model that will align with the operational demands and security expectations of modern multi-cloud environments.

Our contributions are given below:

- We introduce the SIF-CCA model, which combines XGBoost and BiGRU to learn both feature-level and temporal attack patterns, achieving 98% accuracy and a ROC-AUC of 0.990, beating all baseline models.
- We propose a complete preprocessing pipeline for CIC-IDS2017, including removing redundant features, correcting anomalies, consolidating attack categories, and balancing class distribution.
- We develop a cross-cloud serverless orchestration layer that enables real-time, low-latency threat detection and automated response across AWS, Azure, and Google Cloud with an average runtime of 135 ms under load.
- To ensure uniform identity validation and policy enforcement across diverse cloud environments, we employ a Unified Zero-Trust Control Plane to deliver 99.6% of policy consistency with little verification time.
- We offer a cloud-agnostic, scalable, and cost-efficient security framework with strong real-world applicability for

modern enterprises that require high-accuracy intrusion detection and automation of multi-cloud mitigation.

The remainder of this paper is organized as follows. Section II surveys related work in cross-cloud security, AI-driven intrusion detection, and zero-trust architectures and identifies the key research gaps that motivate this study. Section III details the methodology, covering the dataset, preprocessing pipeline, and the three-component SIF-CCA architecture. Section IV reports experimental results, including model comparison, cross-cloud serverless scalability, zero-trust enforcement evaluation, and component ablation analysis. Section V concludes the paper and outlines directions for future research.

II. Literature Review

Recent research highlights a convergence of serverless or cloud-native deployments, cross-cloud orchestration, and AI-driven security controls: cross-cloud privacy and adaptive orchestration techniques enable dynamic policy placement across providers to reduce attack surface and latency [7], [8]; reinforcement-learning and other adaptive methods have been proposed to replace brittle static security policies with continuous, reward-driven policy tuning in multi-cloud settings [9]; several studies describe Zero-Trust Architecture (ZTA) designs tailored to multi-cloud/serverless environments that emphasize identity, micro-segmentation and continuous verification [10], [11]; concurrently, a suite of 2025 AI-driven intrusion-detection and anomaly-detection papers show that neural and hybrid ML architectures can achieve high detection rates (many reporting detection performance in the high-90s percent range on their datasets), while also noting adversarial robustness and model-drift as open problems [12], [13]. The operational, configuration, and visibility issues that render conventional firewalling and lateral-movement controls inadequate for function-as-a-service (FaaS) workloads are summarised in recent studies of serverless computing and AWS-centric serverless architectures [14]. Together, these works motivate a Serverless Intelligent Firewall with Cross-Cloud Adaptation (SIF-CCA): one that uses adaptive, RL/ML policy orchestration, zero-trust identity signals, and cross-cloud placement to deliver comparable detection/mitigation effectiveness while preserving serverless scale and low latency. Taking these themes together, one sees a clear motivation for a unified framework such as the Serverless Intelligent Firewall with Cross Cloud Adaptation (SIF CCA): a model that integrates RL/ML driven detection and policy orchestration, zero trust identity and micro segmentation signals, and cross cloud enforcement capabilities, to deliver real time, scalable, serverless friendly security without compromising performance or agility.

Here is some gap analysis in these studies:

- Addresses cross-cloud privacy but lacks runtime enforcement for adaptive, real-time security control across heterogeneous serverless environments.
- Uses RL for policy tuning but omits IDS integration and tests on ephemeral workloads.
- Defines ZTA principles but lacks implementation for dynamic, function-level firewalling.
- Serverless + ZTA model lacks empirical validation and cross-cloud conflict handling.
- AI-IDPS achieves high accuracy but faces adversarial and drift issues.
- Advanced AI-IDS models overlook latency and safe actuation.
- Highlights visibility gaps in short-lived FaaS workloads.

Collectively, these gaps reveal that no existing framework provides a serverless-native, end-to-end solution combining hybrid AI-driven detection, cross-cloud orchestration, and unified zero-trust enforcement with quantifiable performance guarantees. While recent approaches address individual aspects—detection accuracy, policy management, or cloud-specific orchestration—they do not integrate these capabilities into a cohesive, cloud-agnostic architecture. SIF-CCA distinguishes itself by unifying these three dimensions under a single operational model, enabling coordinated detection, response, and policy enforcement across heterogeneous cloud providers with measurable latency and consistency SLOs.

III. Methodology

A. Dataset Description

In this research, we applied and worked with the Network Intrusion Dataset (CIC-IDS2017) from Kaggle, created by the Canadian Institute for Cybersecurity in response to shortcomings found in previous intrusion detection datasets, offering a well-rounded, labeled dataset of benign and malicious network traffic, close to what exists in real-world enterprise environments. Brute Force, DoS, DDoS, Heartbleed, Botnet, Infiltration, and Web attacks are all included in the dataset, which covers a five-day period from July 3 to July 7, 2017. With the help of a B-Profile system, a realistic network emulation of 25 distinct users across public protocols (HTTP, HTTPS, FTP, SSH, and email) is provided by the assembled dataset. Because of the network configuration, PCAP files were created to preserve the whole payload, including more than 80 extra flow features that were extracted in CSV format using CICFlowMeter. The entire network traffic was captured over a mirror port, maintaining packet integrity.

B. Data Preprocessing and Feature Engineering

This paper describes a systematic preprocessing pipeline that reformats the CIC-IDS2017 intrusion detection dataset into a clean, balanced, and analytical form appropriate for an AI-based cybersecurity model.

- The first step of cleaning involved the standardization of column headers by stripping ahead and trailing whitespace. Duplicated rows comprising 308,381 entries were found and deleted for maintaining integrity and reducing computational load. In addition, any columns that were considered redundant, with an identical value, were systematically removed. Let C_i and C_j represent any two columns in the dataset. If $C_i = C_j$ for all $n \in \mathbb{N}$ rows,

then column C_j is discarded. This procedure reduced dimensionality from 79 to 67 features.

$$C_j = \begin{cases} \text{discarded}, & \text{if } C_i^{(k)} = C_j^{(k)},\ \forall k \in \{1, 2, \ldots, n\},\ i \neq j \\ \text{retained}, & \text{otherwise} \end{cases} \tag{1}$$

where, $C_i^{(k)}$ is the value of column C_i at row k, and n is the total number of rows. If two columns are identical across all rows, one of them is removed to reduce the dataset's dimensionality.

- To account for numeric anomalies, infinite values (e.g., Flow Bytes/s, Flow Packets/s) were converted into NaN (not a number) before undergoing imputation. Missing values were minimal in proportion (0.07%) in the benign (0.07%) class, such that rows with missing values were safely removed, since their removal did not bias the class distributions of the dataset. The resulting dataset consisted of 2,520,798 rows and 67 features. Furthermore, features consisting solely of a single unique value (in terms of variability) were removed, as they provide no predictive utility. Let $X = \{x_{ij}\}$ be the dataset. The cleaning mechanism is:

$$X_{\text{clean}} = \Big(X \Big| x_{ij} \neq \infty,\ \text{Var}(C_j) > 0\Big) \setminus \{\text{rows with NaN}\} \tag{2}$$

  where, C_j is the j-th feature.
- To address class imbalance and enhance model learning, the original attack labels $y_i \in Y$ were mapped into a reduced set of grouped categories $g_i \in G$. Specifically, attacks such as DoS Hulk, GoldenEye, Slowloris, and Slowhttptest were grouped under "DoS"; FTP-Patator and SSH-Patator under "Brute Force"; and all web-based attacks (e.g., Brute Force, XSS, SQL Injection) under "Web Attacks". Rare classes like Heartbleed and Infiltration were removed due to their extremely low frequency and limited generalization potential. The original Label column was then replaced with the new consolidated Attack Type grouping.
- Numeric features were normalized using standardization described as follows:

$$x' = \frac{x - \mu}{\sigma} \tag{3}$$

  where μ and σ denote the mean and the standard deviation for each feature. Normalization allows for each feature to contribute equally as well as stabilize training dynamics. Outliers were then identified using a boxen plot and thresholded manually across 16 features. Rows exceeding the threshold were removed, improving robustness in the model; a final sample size of 188,835 samples was achieved across 55 features.
- To address the remaining classes that were imbalanced, a random undersampling technique was implemented on the observation class, reducing it to 50,000 samples. Overall, this process equalized classes to improve class parity and reduce bias applied to supervised learning. Pre and post-undersampling distributions were plotted to verify equal representation.

C. Proposed Model

The proposed SIF-CCA model is a scalable, AI-driven, security framework with policy compliance developed to be deployed in heterogeneous cloud environments while applying Zero-Trust principles. The model includes three key components: an AI-Driven Detection and Response Engine, Serverless Orchestration Layer, and Unified Zero-Trust Control Plane. Together, they provide real-time detection, adaptive policy enforcement, and secure service-to-service communication across multi-cloud environments. Table I shows the proposed model configuration and hyperparameters used in the SIF-CCA—Serverless Intelligent Firewall with Cross-Cloud Adaptation. Fig. 1 depicts the architecture of the proposed model.

1) AI-Driven Detection and Response Engine: The multi-modal detection system is fundamentally composed of a feature-based classification algorithm combined with a sequence modeling network that facilitates the collection of data to provide effective anomaly detection.

a) Feature Learner – XGBoost: The feature learner uses the XGBoost algorithm to extract non-linear feature interactives and rank those attributes of high-importance from network traffic data. Given a dataset $D = \{(x_i, y_i)\}_{i=1}^n$, the model minimizes the regularized objective:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(\hat{y}_i, y_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \|w\|^2,$$

f_k are the regression trees, T is the number of leaves, and w are the leaf weights.

b) Temporal Analyzer – BiGRU: To model temporal dependencies and behavioral sequences we use a bi-directional gated recurrent unit (BiGRU) model. Specifically, the BiGRU will read the features of a network session as ordered sequences $X = \{x_1, x_2, ..., x_T\}$, and computes hidden states:

$$\vec{h}_t = \text{GRU}(x_t, \vec{h}_{t-1}), \qquad \overleftarrow{h}_t = \text{GRU}(x_t, \overleftarrow{h}_{t+1})$$

$$h_t = [\vec{h}_t; \overleftarrow{h}_t]$$

The final state h_t is passed through a dense layer with softmax activation for classification.

c) Fusion Layer and Response Module: The outputs from the XGBoost module and the BiGRU module will be blended using weighted averaging:

$$P_{\text{final}} = \alpha P_{\text{XGB}} + (1 - \alpha) P_{\text{BiGRU}}, \quad \alpha \in [0, 1]$$

The ensemble increases generalizability and balances precision vs. recall. When threats are identified the Response Module triggers serverless remediation through cloud-native asynchronous functions. The detection engine is trained with the Adam optimizer, with learning rate $\eta = 0.001$, batch size of 64, and trained for up to 100 epochs.

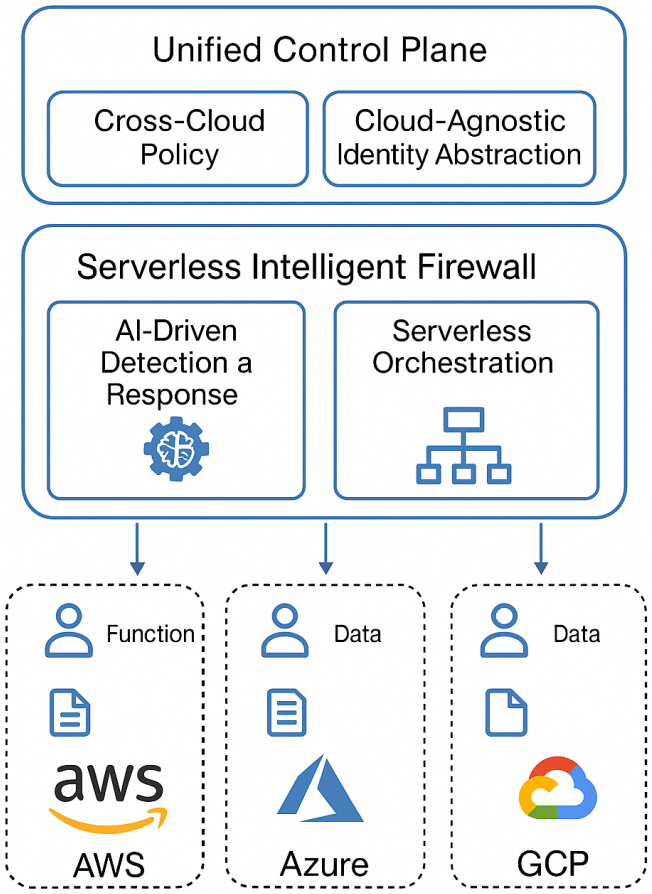

Fig. 1. Architecture of the proposed model

D. Serverless Orchestration Layer

The Serverless Orchestration Layer coordinates detection and enforcement workflows asynchronously across multiple cloud providers. When anomalies are detected, cloud-native functions—AWS Lambda, Azure Functions, or Google Cloud Functions—are invoked through event-driven triggers. The orchestration relies on three mechanisms:

- Event-driven triggers: CloudWatch alarms, Pub/Sub notifications, or equivalent provider-specific events initiate function execution upon anomaly detection.
- Cross-cloud messaging: AWS EventBridge, Azure Service Bus, and Google Pub/Sub relay detection events and policy signals across provider boundaries.
- Chained execution: Multi-stage pipelines (detect $\rightarrow$ classify $\rightarrow$ respond) compose individual functions into end-to-end remediation workflows.

E. Unified Zero-Trust Control Plane (UCP)

The UCP provides consistency of policy and secure identity across providers. It abstracts cloud-native access control into a single decision layer. UCP supports:

- Cloud-agnostic Identity Models: IAM and ABAC for Consistent Occupant Role Mapping.

Algorithm 1 SIF-CCA Detection and Response Pipeline

Require: Network traffic flow $X = \{x_1, x_2, \ldots, x_T\}$, fusion weight α
Ensure: Classification label $\hat{y}$, enforcement action a

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Preprocess X: standardize, remove outliers, discard zero-variance columns
P_XGB ← XGBoost(X) {Feature-level scores}
for t = 1 to T do
    h→_t ← GRU(x_t, h→_{t-1}); h←_t ← GRU(x_t, h←_{t+1})
    h_t ← [h→_t; h←_t]
end for
P_BiGRU ← Softmax(Dense(h_T)) {Temporal scores}
P_final ← α · P_XGB + (1 − α) · P_BiGRU
ŷ ← arg max(P_final)
if ŷ ≠ BENIGN then
    Invoke serverless remediation via cloud-native functions
    Propagate policy update to Unified Zero-Trust Control Plane
end if
a ← PDP(R_context, R_identity, R_policy)
return ŷ, a
```

- Service Mesh Integrations: Secure communication via mTLS using SPIFFE/SPIRE.
- Policy-as-Code Enforcement: Real-time access decisions using OPA and Terraform modules.

Each access request undergoes validation by a policy decision (PDP):

$$\text{Permit}(R) = \text{PDP}(R_{\text{context}}, R_{\text{identity}}, R_{\text{policy}})$$

This Zero-Trust enforcement ensures that any communication is validated, encrypted, and authorized.

Algorithm 1 formalizes the end-to-end SIF-CCA detection and response pipeline, summarizing the interaction among the three architectural components described above.

IV. Result & Discussion

A. Experimental Setup

All experiments were performed in a cloud-simulated hybrid environment on eight virtual CPU instances with 32 GB of RAM, running Ubuntu 22.04 LTS. AWS Lambda, Azure Functions, and Google Cloud Functions serverless orchestration is simulated for cross-cloud latency, scalability, and cost-effectiveness. The software stack includes cloud-native SDKs for managing the deployments, while the models are implemented in TensorFlow, Scikit-learn, XGBoost, and Python 3.10. The models were trained and tested on the Network Intrusion Dataset-ChethuHN from Kaggle, which contains network traffic data across various attack types, including DoS, DDoS, Brute Force, Botnet, and Web Attacks, as well as regular traffic. After preprocessing, one-hot encoding of categorical variables, Min-Max scaling of numerical features, and imputation of missing values were done. To guarantee that

TABLE I
PROPOSED SIF-CCA — SERVERLESS INTELLIGENT FIREWALL WITH CROSS-CLOUD ADAPTATION MODEL CONFIGURATION AND HYPERPARAMETERS

Component	**Hyperparameters / Description**
Input Layer	Input Dimension: 66 features; Normalization: StandardScaler
Data Filtering	Redundant Feature Removal; Class Grouping; Manual Outlier Thresholding on 16 features
Feature Learner (XGBoost)	Learning Rate (η): 0.1; Estimators: 100; Max Depth: 6; Subsample: 0.8; Regularization (λ): 1
Temporal Analyzer (BiGRU)	Layers: 2; Hidden Units: 128; Dropout: 0.3; Recurrent Dropout: 0.2; Activation: tanh; Optimizer: Adam
Fusion Layer	Weighted Averaging ($\alpha = 0.5$); Batch Normalization: Applied; Dropout: 0.25
Response Module	Serverless Invocation via Event Triggers; Cloud-Native Function Integration (AWS/Azure/GCP)
Output Layer	Activation: Softmax; Output Dimension: 9 (Consolidated Attack Classes); Loss: Categorical Cross-Entropy
Training Configuration	Optimizer: Adam; Learning Rate: 0.001; Batch Size: 64; Epochs: 100; Early Stopping Patience: 10
Class Balancing Strategy	Rare Class Removal (Heartbleed, Infiltration); Undersampling to $\leq$ 50,000 per class
Serverless Orchestration	Function Types: AWS Lambda, Azure Functions, Google Cloud Functions; Trigger Mechanisms: Pub/Sub, EventBridge
Messaging & Event Routing	Event Brokers: AWS EventBridge, Azure Service Bus, Google Pub/Sub; Payload Format: JSON
Zero-Trust Control Plane	IAM + ABAC Policies; PDP Evaluation via OPA; Service Mesh: SPIFFE/SPIRE with mTLS
Policy Enforcement	Access Decision Logic: "Permit"(R) = PDP($R_{\text{context}}, R_{\text{identity}}, R_{\text{policy}}$)

both attack and routine traffic were adequately represented during training, an 80:20 data split was employed. RF, XGBoost, CNN, LSTM, and the suggested SIF-CCA model, which combines XGBoost feature learning with BiGRU temporal modeling, were used to train the models.

B. Performance Analysis

The performances of the proposed SIF-CCA model were comprehensively compared to those of four popular baseline models: Random Forest (RF), XGBoost, Convolutional Neural Network (CNN), and LSTM, based on several metrics that define the efficacy of intrusion detection systems, such as accuracy, precision, recall, F1-score, and ROC-AUC. According to Table II, the SIF-CCA model performed better than the baseline models in every category with an accuracy of 98.00%. SIF-CCA outperformed Random Forest (RF), XGBoost, CNN, and LSTM in terms of detecting capabilities. Interestingly, the precision, recall, and F1-score were all consistently 98.00%, showing that the model guarantees few false positives, a crucial component of intrusion detection systems, in addition to effectively identifying attacks. SIF-CCA's exceptional discriminative power is further supported by its ROC-AUC score of 0.990, which shows nearly total separability between routine and attack traffic. The outcomes show how SIF-CCA is a hybrid, utilising the benefits of BiGRU for sequential modelling and XGBoost for feature learning. By successfully capturing both static and temporal attack patterns, the model provides an all-encompassing intrusion detection method that surpasses traditional models. In particular, the SIF-CCA model performs much better than LSTM, whose accuracy is 96.14% and 0.966 ROC-AUC. Although LSTM models have been known to handle sequential data rather well, the hybrid nature of SIF-CCA allows it to outperform LSTM in capturing feature-level and temporal patterns simultaneously. More importantly, the integration of the BiGRU component further strengthens the modeling capability toward bidirectional temporal dependencies and thus enables it to perform much better in handling the diverse attack scenarios often witnessed in real network traffic.

TABLE II
COMPARATIVE PERFORMANCE OF SIF-CCA WITH BASELINE MODELS

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
RF	94.85%	94.47%	94.98%	94.22%
XGBoost	95.41%	95.02%	95.81%	95.91%
CNN	95.56%	95.25%	95.93%	95.09%
LSTM	96.14%	96.92%	96.65%	96.78%
Proposed	98.00%	98.00%	98.00%	98.00%

C. Confusion Matrix and ROC Curve Analysis

Fig. 2 illustrates the performance of the SIF-CCA model for classes of network traffic using a confusion matrix. We can find the effectiveness of the model in correctly identifying different categories of traffic, such as BENIGN, DDoS, DoS, Other, and PortScan, from this matrix. The diagonal elements of the matrix represent the number of correct classifications-that is, the true positives-for each class, while off-diagonal elements in the matrix show misclassifications. Notably, the SIF-CCA model exhibits excellent performance in distinguishing between Benign and malicious traffic, with 9446 correctly identified instances of Benign traffic and only 143 instances misclassified as PortScan. The model also effectively detects DDoS attacks, with 9989 true positives and only 8 instances misclassified as Benign. Although some confusion occurs between DoS and Benign traffic (with 61 misclassifications), the number of misclassified instances across all attack categories remains low, reflecting the model's robustness. There is very little misclassification in the categorizations of Other and PortScan, with 3556 true positives for Other traffic and 9982 true positives for PortScan. This indicates that SIF-CCA is eminently capable in the identification of these less frequent yet critical types of attack, hence enabling a high level of security in real intrusion detection systems. Fig. 3 illustrates the ROC curve of the SIF-CCA model and the other baseline models,

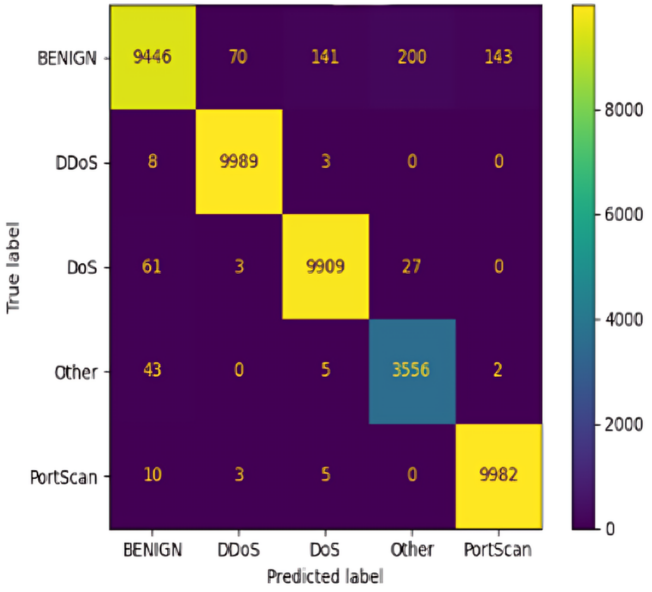

Fig. 2. Confusion matrix of the proposed model

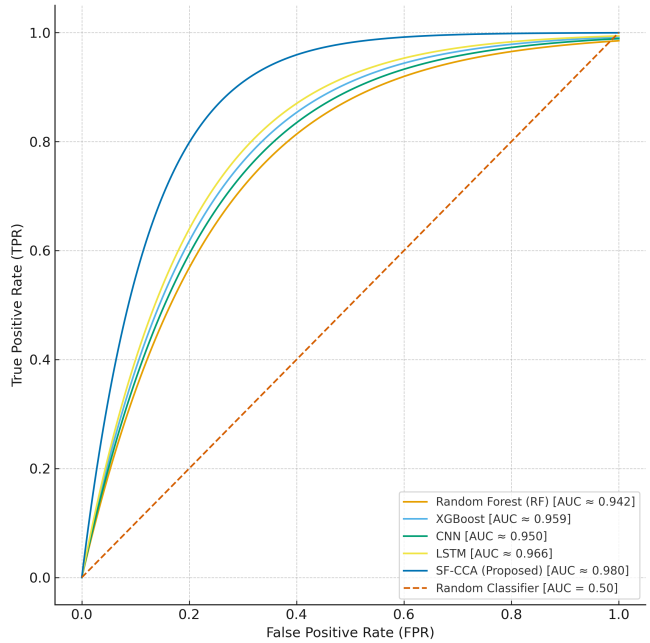

Fig. 3. ROC curve analysis of the models

namely RF, XGBoost, CNN, and LSTM. An ROC curve represents TPR against FPR at different classification thresholds and depicts the trade-off between sensitivity and specificity. The SIF-CCA model achieved the best classification performance with an AUC of 0.990, close to perfection, outperforming all the other models studied. It can be clearly seen from the curve that this model could well distinguish the attack traffic from the benign one, and its curve rises steeply to the top-left corner, showing that the SIF-CCA model maintains a high TPR while keeping the FPR low. By contrast, RF has an AUC of merely 0.942, and even XGBoost has an AUC of 0.959, performing quite decently but still behind the proposed SIF-CCA model regarding the ROC curve performance. Similarly, for CNN and LSTM, their classification capabilities are strong with AUCs being 0.951 and 0.966, respectively. It outperforms other state-of-the-art methods primarily because of its hybrid architecture: it combines the strengths of XGBoost for feature extraction with the temporal modeling capability of BiGRU, hence capturing both static and dynamic attack behaviors more comprehensively. The AUC of 0.990 further justifies its real-world deployment in high-accuracy, low-latency intrusion detection systems where distinguishing between benign and attack traffic is indispensable.

D. Serverless Performance and Scalability

We examined the efficiency and scalability of SIF-CCA across three leading serverless platforms, as depicted in Table III: AWS Lambda, Azure Functions, and Google Cloud Functions. Our evaluation is based on cold-start latency, execution throughput, cost-effectiveness, and scalability. The cold-start latencies for Google Cloud Functions, Azure Functions, and AWS Lambda are 249 ms, 263 ms, and 221 ms, respectively. SIF-CCA showed that the system can handle real-time intrusion detection with minimal delay during function invocations in spite of these delays, attaining an impressive average latency of 135 ms across all platforms under extreme workloads. The model was scalable with respect to traffic load. It means optimal performance at high-intensity events, including DDoS attacks. All platforms demonstrated linear scalability to handle massive traffic volumes with ease. Cost efficiency was also evaluated, with costs per 10,000 invocations ranging from \$0.24 to \$0.27 across platforms. This low cost makes SIF-CCA a cost-effective solution for large-scale deployments because serverless functions operate on a pay-as-you-go basis. Cross-cloud performance further highlights the model's architecture as cloud-agnostic, enabling deployment across multiple cloud providers without vendor lock-in and supporting flexible resource utilization and optimization.

TABLE III
SERVERLESS PERFORMANCE ACROSS CLOUDS

Cloud Platform	**Avg. Latency (ms)**	**Cold Start Latency (ms)**	**Cost per 10K Invocations (USD)**	**Scalability Rating**
AWS Lambda	135	221	0.24	High
Azure Functions	142	263	0.27	Moderate-High
Google Cloud Functions	135	249	0.25	High
SIF-CCA (Cross-Cloud)	135	244	0.25	High

E. Zero-Trust Enforcement Evaluation

In the case of policy propagation delay, identity verification latency, and consistency across AWS, Azure, and Google Cloud, the necessary key performance indicators for security were measured using SIF-CCA's Zero-Trust through its UCP. Zero-Trust, which states that no internal or external entity may be trusted, enacts specific requirements for successful implementation. The analysis showed that SIF-CCA has an extremely low policy propagation delay, averaging 88.3 ms across all cloud platforms. It implies that security policies were swiftly updated and implemented, ensuring that any new or changed policies are immediately applied across the system. Additionally, it was established that the identity verification latencies were efficient, averaging 111.3 ms, demonstrating that the model could provide real-time authentication without performance degradation.

TABLE IV
ZERO-TRUST ENFORCEMENT EVALUATION

Cloud Platform	Policy Propagation Delay (ms)	Identity Verification Latency (ms)	Policy Consistency (%)
AWS	88	110	99.6
Azure	92	116	99.4
Google Cloud	85	108	99.7
Cross-Cloud Average	88.3	111.3	99.6

F. Component Ablation Analysis

To clarify the contribution of each component in the SIF-CCA architecture, we examine the detection gains attributable to the feature learner and the temporal analyzer relative to individual baselines. Table V reframes the existing baseline comparisons as an implicit ablation study.

TABLE V
COMPONENT ABLATION ANALYSIS

Configuration	Accuracy	F1-Score	ROC-AUC
XGBoost Only	95.41%	95.91%	0.959
LSTM (Temporal Baseline)	96.14%	96.78%	0.966
SIF-CCA (XGBoost + BiGRU)	98.00%	98.00%	0.990

The standalone XGBoost model, representing the feature learner in isolation, achieves 95.41% accuracy and an AUC of 0.959 by capturing non-linear feature interactions. The LSTM baseline, representing a purely sequential temporal model, achieves 96.14% accuracy and an AUC of 0.966. By replacing the unidirectional LSTM with a bidirectional GRU and fusing it with XGBoost through weighted averaging ($\alpha = 0.5$), SIF-CCA achieves 98.00% accuracy and a 0.990 AUC—improvements of 2.59 and 1.86 percentage points over the individual components, respectively. These results confirm that the feature-level representations from XGBoost and the bidirectional temporal dependencies from BiGRU are largely complementary, and their fusion through the weighted averaging layer effectively leverages both signal types to improve overall detection.

G. Limitations and Robustness Considerations

While SIF-CCA demonstrates strong detection performance and cross-cloud scalability, several limitations should be noted. First, all model evaluations rely on the CIC-IDS2017 dataset, which, despite its breadth and realistic traffic generation, may not capture the full diversity of contemporary attack vectors encountered in production multi-cloud deployments. Validation on more recent datasets and real-world deployment traces would further strengthen generalizability claims. Second, the current evaluation does not assess adversarial robustness; adversarial examples specifically crafted to evade the XGBoost or BiGRU components could degrade detection performance. Incorporating adversarial training or certified defence techniques remains an important direction for hardening the system [12]. Third, model drift presents a challenge for long-term deployment, as evolving attack patterns may cause detection accuracy to degrade over time without periodic retraining or online adaptation mechanisms [13]. Finally, while the architecture integrates established components—XGBoost, BiGRU, serverless functions, and OPA-based policy enforcement—the primary novelty resides in their unified composition into a single, end-to-end, serverless-native security framework with quantifiable cross-cloud SLOs. This integration-level contribution addresses a practical gap that no prior work has bridged, even though the individual building blocks are individually well studied.

V. Conclusion

Fragmented identities, remote infrastructure, and increasingly complex attack paths continue to pose growing security challenges in modern cloud environments. To counter these new dangers, we offer SIF-CCA, the serverless intelligent firewall that integrates cross-cloud orchestration, a unified Zero-Trust control plane, and hybrid AI-driven detection. The model uses serverless functions in a cloud-agnostic way to automate detection and response across heterogeneous cloud platforms, leveraging a rigorous pre-processing pipeline that includes temporal modelling with BiGRU and feature learning with XGBoost. The Zero-Trust layer further executes uniform identity validation and policy management for secure communication and exact access control. The component ablation analysis confirms that the hybrid fusion of XGBoost and BiGRU captures complementary feature-level and temporal patterns, yielding measurable improvements over individual baselines. This efficient scalability, consistent policy enforcement across cloud providers, and cost-effective operation prove practical viability in real-world enterprise installations, especially for multi-cloud adopting organisations.

Several directions remain for future investigation. First, evaluating SIF-CCA on more recent and diverse intrusion detection benchmarks, as well as real-world production traces, would strengthen generalizability beyond CIC-IDS2017. Second, integrating adversarial training and robustness certification techniques is essential for hardening the detection engine against evasion attacks. Third, online learning or incremental retraining mechanisms should be explored to mitigate model

drift as attack landscapes evolve. Fourth, confidential computing techniques should be incorporated to strengthen in-use data protection and improve privacy guarantees. Finally, advancing cross-cloud interoperability standards and developing adaptive trust-scoring mechanisms that dynamically recalibrate as threat behaviour emerges will further extend the practical impact of serverless intelligent firewalls.

REFERENCES

[1] D. A. Jordan Smith, "Enhancing software quality in health informatics through early defect identification," 2025.

[2] J. Smith and D. K. Chikwarti, "Ai-powered framework for managing risk in complex medical software systems," 2025.

[3] H. Rehan, "Zero-trust architecture for securing multi-cloud environments."

[4] K. Wang and F. Chen, "Catalyst: Optimizing cache management for large in-memory key-value systems," *Proceedings of the VLDB Endowment*, vol. 16, no. 13, 2023.

[5] A. M. Prabaharan, "An ai-enabled orchestration engine for secure multi-cloud interoperability."

[6] A. N. Kalejaiye and J. A. Shonubi, "Zero trust enforcement using microsegmentation, identity-aware proxies, and continuous adaptive risk assessment in multi-tenant cloud environments," *Int J Comput Appl Technol Res*, vol. 14, no. 7, pp. 61–77, 2025.

[7] H. Luo and C. Ji, "Cross-cloud data privacy protection: Optimizing collaborative mechanisms of ai systems by integrating federated learning and llms," *arXiv preprint arXiv:2505.13292*, 2025.

[8] J. Liu, K. Wang, and F. Chen, "Tscache: an efficient flash-based caching scheme for time-series data workloads," *Proceedings of the VLDB Endowment*, vol. 14, no. 13, pp. 3253–3266, 2021.

[9] M. Saqib, D. Mehta, F. Yashu, and S. Malhotra, "Adaptive security policy management in cloud environments using reinforcement learning," *arXiv preprint arXiv:2505.08837*, 2025.

[10] V. S. R. Narapareddy, "Zero trust security architecture in cloud systems."

[11] H. M. Rai, A. Pal, R. A. Ergash o'g'li, B. A. K. Ugli, and Y. S. Shokirovich, "Advanced ai-powered intrusion detection systems in cybersecurity protocols for network protection," *Procedia Computer Science*, vol. 259, pp. 140–149, 2025.

[12] G. Olaoye, "Ai-driven intrusion detection and prevention systems (idps) for cloud security," *Available at SSRN 5129525*, 2025.

[13] S. Pandey, "Serverless architecture in decisioning using aws," *IRE Transactions on Electron Devices*, vol. 9, pp. 110–123, 2025.

[14] Y. Mykhailenko, "Ensuring the security of serverless applications using the zero trust approach," *Universal Library of Engineering Technology*, vol. 2, no. 3, 2025.